Starter

For the following questions,

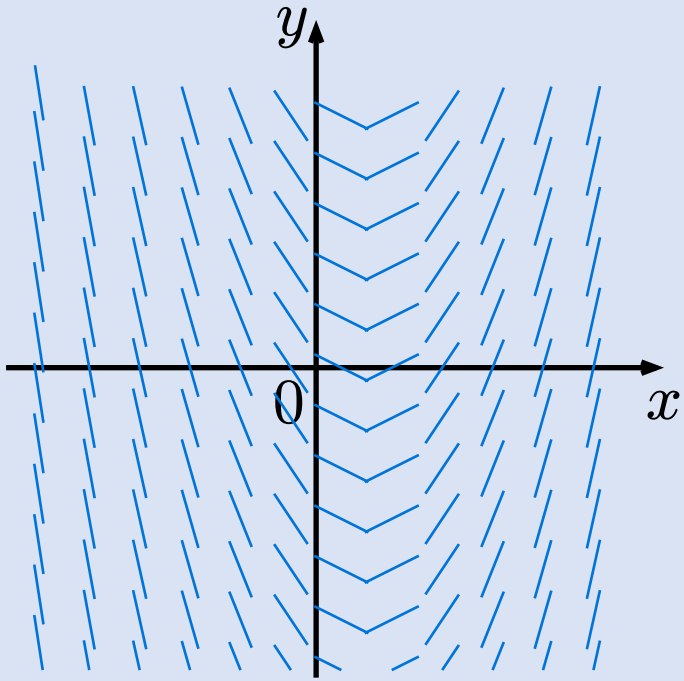
1. Sketch the gradient field for $-3 \leq x \leq 3, -3 \leq y \leq 3$
2. Solve the DE with the given initial conditions. Sketch your solution on your gradient field.

a.
$$\frac{\mathrm{d}y}{\mathrm{d}x} = 2x - 1, \text{initial:}(1,1)$$

b.
$$\frac{\mathrm{d}y}{\mathrm{d}x} - 2xy = 0, \text{initial: } (0,1)$$

c.
$$\tanh(x)\frac{\mathrm{d}y}{\mathrm{d}x} + \operatorname{sech}^2(x)y = 1, \text{initial: } (1,\ \coth(1))$$

Solutions



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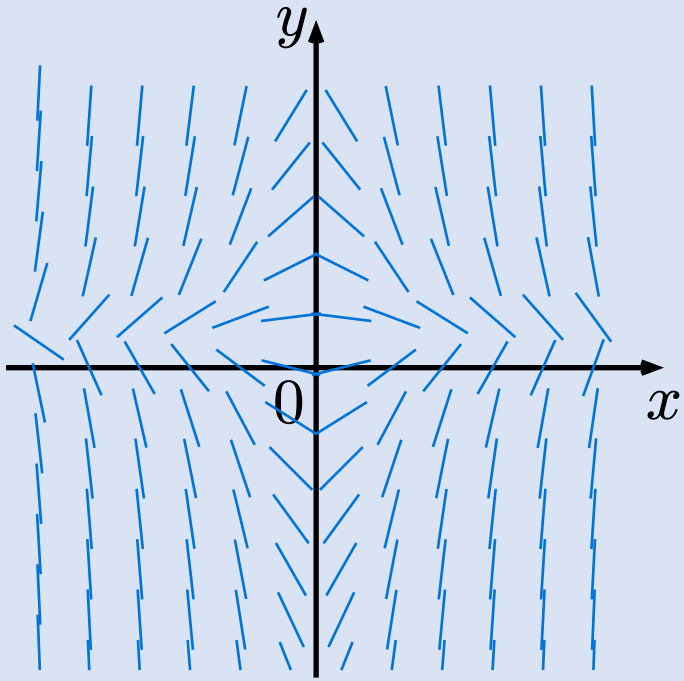

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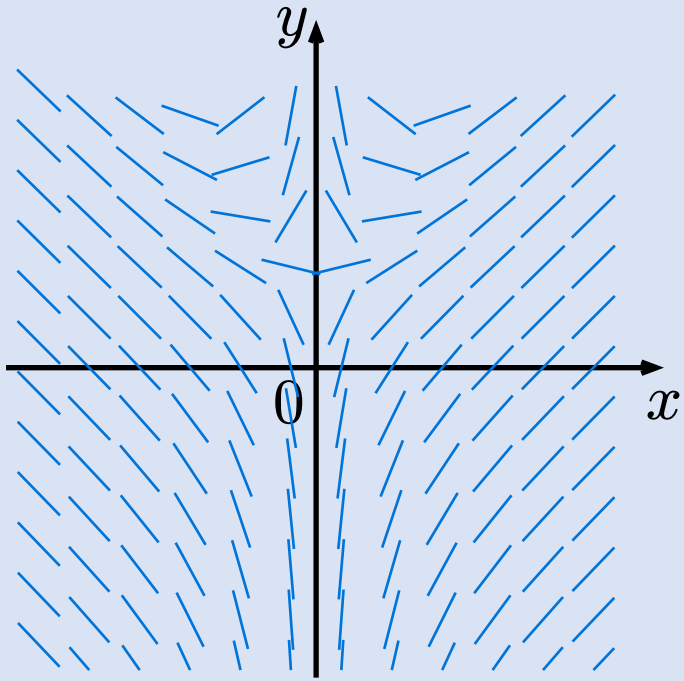

Find the general solution

1. $\dfrac{\mathrm{d}y}{\mathrm{d}x} = x - 3xy$
2. $\dfrac{\mathrm{d}y}{\mathrm{d}x} = x - 3y$

Why is that second one different if it looks so similar?

Find the general solution

1. $\dfrac{\mathrm{d}y}{\mathrm{d}x} = x - 3xy \quad \Rightarrow y = Ae^{-\frac{3}{2}x^2} + \dfrac{1}{3}$
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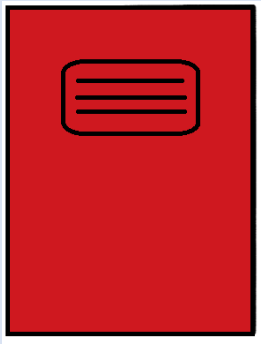
Why is that second one different if it looks so similar?

We'll be looking at these questions this lesson. The variables cannot be separated, and it's not immediately a product rule.

Rearranging DEs



The standard way to write a DE is with the y parts on one side, and everything else on the other.

EG:

$$\frac{\mathrm{d}y}{\mathrm{d}x} = xy + x + 1 \rightarrow \frac{\mathrm{d}y}{\mathrm{d}x} - xy = x + 1$$

This does help us to spot those product-rule questions too.

Classifying DEs

Different types of DE will have different typical solutions and methods. Classifying them is then really useful!

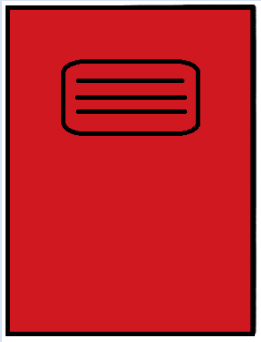
Order of a DE

$$\frac{\mathrm{d}y}{\mathrm{d}x} = x - 3xy \rightarrow \text{1st order}$$

$$\frac{\mathrm{d}^2y}{\mathrm{d}x^2} + 2\frac{\mathrm{d}y}{\mathrm{d}x} + 4y = x$$

$$\rightarrow \text{2nd order}$$

Classifying DEs



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Homogeneous vs non-homogeneous

$$\frac{\mathrm{d}y}{\mathrm{d}x} = 3xy \rightarrow \text{Hom.}$$

$$\frac{\mathrm{d}y}{\mathrm{d}x} = x + y \rightarrow \text{non-Hom.}$$

Classifying DEs

Linear vs non-linear

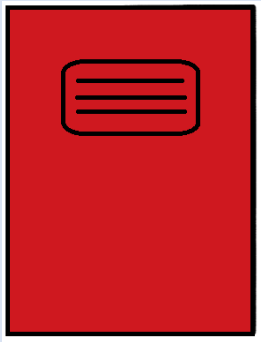
$$\frac{\mathrm{d}y}{\mathrm{d}x} = 3xy + x^2 \rightarrow \text{linear}$$

$$2xy\frac{\mathrm{d}y}{\mathrm{d}x} + y^2 = x \rightarrow \text{non-linear}$$

Classifying DEs

- **Order**: Like a polynomial, but looking at the highest order of Differential.

Classifying DEs



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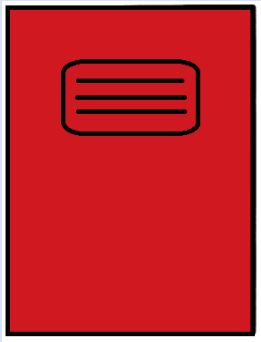

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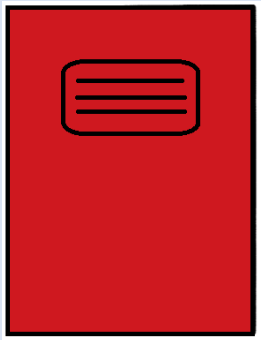

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- **Linear**: Any equation where terms are only involving y and derivatives of y.

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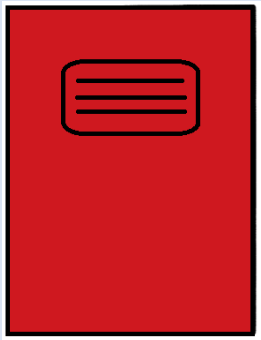

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- **Linear**: Any equation where terms are only involving y and derivatives of y.
- **Non-linear**: An equation where there are terms involving non-linear y terms (eg. $y^2, e^y, y\frac{\mathrm{d}y}{\mathrm{d}x}$)

So far

We've seen 1st order homogenous DEs that can be solved by separation of variables. (linear and non-linear)
EG:

$$\frac{\mathrm{d}y}{\mathrm{d}x} - 2xy = 0 \text{ or } \frac{\mathrm{d}y}{\mathrm{d}x} = e^x y^2$$

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We've seen 1st order non-linear non-homogeneous equations where the product rule applies,
EG:

$$\sin x \frac{\mathrm{d}y}{\mathrm{d}x} + y \cos x = 2x$$

The next step

The next kind of differential equation we're going to tackle is when the product rule seems like it might apply,
but the terms don't quite line up.

$$x^2 \frac{\mathrm{d}y}{\mathrm{d}x} + 4xy = 3$$

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$$x^2 \frac{\mathrm{d}y}{\mathrm{d}x} + 4xy = 3$$

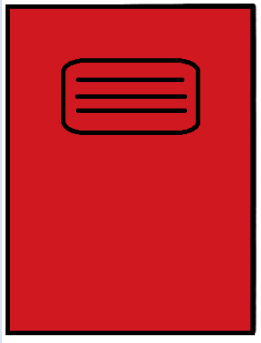
$$\times x^2 \qquad x^4 \frac{\mathrm{d}y}{\mathrm{d}x} + 4x^3 y = 3x^2 \qquad \times x^2$$

$$\frac{\mathrm{d}}{\mathrm{d}x}(x^4 y) = 3x^2$$

$$x^4 y = x^3 + c$$

$$y = \frac{1}{x} + \frac{c}{x^4}$$

The next step



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$$x^2 \frac{\mathrm{d}y}{\mathrm{d}x} + 4xy = 3$$

We call x^2 here the $\rightarrow$ **integrating factor**

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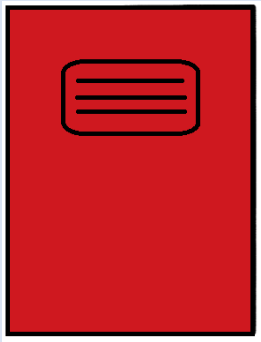


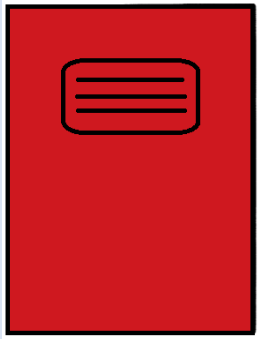
An Integrating factor can be used to create perfect derivatives for 1st order linear differential equations.
(Although the method can be modified to work with non-linear ones)

Useful for those in the form:

$$\frac{\mathrm{d}y}{\mathrm{d}x} + P(x)y = Q(x)$$

(Linear, in terms of y)

Integrating Factor



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Useful for those in the form:

$$\frac{\mathrm{d}y}{\mathrm{d}x} + P(x)y = Q(x)$$

(Linear, in terms of y)

It can be shown that the integrating factor is always $e^{\int P(x)\mathrm{d}x}$

Find the general solutions:

$$\frac{\mathrm{d}y}{\mathrm{d}x} = x - 3y$$

$$\frac{\mathrm{d}y}{\mathrm{d}x} + \frac{2}{x}y = \frac{4}{x^2}$$

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$$\frac{\mathrm{d}y}{\mathrm{d}x} + 3y = x \to P(x) = 3$$

$$\mathrm{IF} = e^{\int 3\mathrm{d}x} = e^{3x}$$

$$\therefore e^{3x}\frac{\mathrm{d}y}{\mathrm{d}x} + 3e^{3x}y = xe^{3x}$$

$$e^{3x}y = \int xe^{3x}\mathrm{d}x$$

$$\ldots$$

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$$P(x) = \frac{2}{x} \rightarrow \mathrm{IF} = e^{\int \frac{2}{x}\,\mathrm{d}x} = x^2$$

$$\therefore x^2\frac{\mathrm{d}y}{\mathrm{d}x} + 2xy = 4$$

$$\therefore x^2y = \int 4\mathrm{d}x$$

$$\ldots$$

You Try

Find the general solution to:

1. $$x\frac{\mathrm{d}y}{\mathrm{d}x} + 4y = x^{-\frac{5}{2}}$$
2. $$\frac{\mathrm{d}y}{\mathrm{d}x} - 2y = e^{10x}$$
3. $$\frac{\mathrm{d}y}{\mathrm{d}x} + 2xy = \sin x$$
4. $$\frac{\mathrm{d}y}{\mathrm{d}x} + \frac{y}{x} = 2x^3$$
5. $$2x\frac{\mathrm{d}y}{\mathrm{d}x} - y = x^2$$
6. $$\frac{\mathrm{d}y}{\mathrm{d}x} + \frac{2y}{x} = x$$
7. $$\frac{\mathrm{d}y}{\mathrm{d}x} + yx = 4x$$
8. $$t\frac{\mathrm{d}x}{\mathrm{d}t} - 3x = t^2$$

“Exam”ple question

10 **(i)** Obtain the solution to the differential equation

$$x\frac{\mathrm{d}y}{\mathrm{d}x}+3y=\frac{1}{x}\ ,\ \text{where}\ x>0\ ,$$

given that $y=1$ when $x=1$. **[7]**

(ii) Deduce that y decreases as x increases. **[2]**

			$x^3y=\frac{1}{2}x^2+c$	A1A1	1.1 1.1	A1 each side of equation	
			$x=1, y=1 \Rightarrow c=\frac{1}{2}$	B1	1.1	Use condition	
			$y=\frac{1+x^2}{2x^3}$	A1	2.5	Expressing y as a function of x (can be in part **(ii)**)	
				[7]			
10	**(ii)**		$y=\frac{1}{2x^3}+\frac{1}{2x}$	B1	2.2a	Write y as sum of two fractions OR differentiate	
			Explain that y is the sum of two decreasing functions and hence decreasing	E1	2.4	OR show that the derivative is negative hence decreasing	

“Exam”ple question

11 A particular radioactive substance decays over time.

A scientist models the amount of substance, x grams, at time t hours by the differential equation

$$\frac{dx}{dt} + \frac{1}{10}x = e^{-0.1t}\cos t.$$

(i) Solve the differential equation to find the general solution for x in terms of t. **[3]**

Initially there was 10 g of the substance.

(ii) Find the particular solution of the differential equation. **[2]**

(iii) Find to 6 significant figures the amount of substance that would be predicted by the model at

(a) 6 hours, **[1]**

(b) 6.25 hours. **[1]**

(iv) Comment on the appropriateness of the model for predicting the amount of substance over time. **[1]**

“Exam”ple question

Question			Answer	Marks	AO	Guida
11	(i)		$\frac{dx}{dt}+0.1x=\cos t e^{-0.1t} \Rightarrow \text{I.F}=e^{0.1t}$	B1	3.3	
			$\Rightarrow e^{0.1t}\times\frac{dx}{dt}+0.1e^{0.1t}\times x=\cos t$			Multiplying throughout by IF and writing RHS as an exact derivative of a product
			$\Rightarrow \frac{d}{dx}\left(e^{0.1t}x\right)=\cos t$	M1	3.4	
			$\Rightarrow e^{0.1t}x=\sin t+c \Rightarrow x=(\sin t+c)e^{-0.1t}$	A1 [3]	3.4	
	(ii)		$x=(\sin t+c)e^{-0.1t};\ x=10,\ t=0 \Rightarrow c=10$	M1	3.4	Substituting $x=10$ and $t=0$ to find a value for c
			$x=(\sin t+10)e^{-0.1t}$	A1 [2]	3.4	
	(iii)	(a)	5.33477	B1 [1]	3.4	to 6sf or better
	(iii)	(b)	5.33486	B1 [1]	3.4	to 6sf or better
	(iv)		E.g. the amount at 6.25 hours is higher than at 6 hours, so the amount is not always decreasing, and so model is not appropriate (for predicting decay)	B1 [1]	3.5a	Conclusion required